**Candidate's name**: ……

| INDEX NUMBER |
| --- |
| |

**Signature**: ……..

**545/1**
**CHEMISTRY**
**Paper 1**
**(Theory)**
**July, 2025**
**2 hours**

**MATIGO EXAMINATIONS BOARD**

**MOCK EXAMINATIONS**

**Uganda Certificate of Education**

**CHEMISTRY**
**Paper 1**
(Theory)

**2 hours**

## INSTRUCTIONS TO CANDIDATES

*This paper consists of* **two** *sections:* **A** *and* **B**. *It has* **six** *examination items.*

*Section* **A** *has* **two** *compulsory items.*

*Section* **B** *has* **two** *parts;* **I** *and* **II**. *Answer* **one** *item from each part.*

*Answers to section* **A must** *be written in the spaces provided while answers to section* **B must** *be written in the answer booklet(s) provided.*

*Answer* **four** *items in all.*

*Any additional item(s) answered will* **not** *be marked.*

| For official use only | | | | | | | |
| --- | --- | --- | --- | --- | --- | --- | --- |
| **Item No.** | **1** | **2** | **3** | **4** | **5** | **6** | **Total wt. score** |
| **Wt. Score** | | | | | | | |

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## SECTION A

*Attempt all items in this section*

**Item 1**

A teenage school girl who had gone to attend an introduction ceremony got so excited when she saw a beverage known as Smirnoff among many drinks like soda being served. She requested to be served with both drinks although she did not know much about Smirnoff.

**Task**

As a chemistry student;

(a) Identify the category of the drink(s).

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(b) Briefly explain the use of the drink(s) when consumed.

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(c) Explain the impact of excessive consumption of the drink

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(d) Compare the drinks

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**Item 2**

During research, a learner found out that when crude oil is cracked, hydrocarbons of short chains are formed which are greatly used in daily life. He also found out that 20kg of the liquid reactant had been cracked.

$$C_{10}H_{22\,(l)} \longrightarrow C_7H_{16\,(l)} + C_3H_{6\,(g)}$$

The learner did not understand the information and has approached you for guidance.

(H = 1, C = 12)

**Task**:
Help the student to;

(a) appropriately identify with reason, the category of compound.

(b) understand the properties of the gaseous product.

(c) determine the amount of the gaseous product formed

(d) explain the impact of the product compound

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## SECTION B

## PART I

*Answer only* **one** *item from this part. Answers should be written in the answer booklet(s) provided.*

**Item 3**

Aluminium is widely used in construction industry in Uganda. This has resulted into need for the material to meet the demands. In one of the areas in Uganda, it has been found that there are many aluminum deposits. The government has cleared a local investor to set up a factory to produce the material so that he can tap into the opportunity.

The community is puzzled about the process of extracting the mineral together with its effect on the environment and benefit other than employment.

Through school Head Teachers, the investor has organized for Radio and Television talks to sensitize the community and you have been chosen to represent your school.

**Task:**

Make a write up of the presentation you will make.

**Item 4**

Many community members in Karamoja do not have access to hydroelectricity for cooking and lighting. They have been told that the waste from their animals can help them overcome the challenge, however, they have no idea how this can be done, its effect on the environment and benefit other than employment.

You have been contacted by one of the members in the district to take him through what can be done.

**Task**:

Write down the message you would deliver to this person.

## PART II

*Answer only* **one** *item from this part. Answers should be written in the answer booklet(s) provided.*

**Item 5**

Due to rapid population growth and the need for money, some natural resources have been exploited e.g. making of cement, road construction, and building, causing proportional environmental degradation

Your class is spearheading an environmental conservation awareness and preservation campaign in the district on the type of resource. You have been selected as their class speaker.

**Task:**

Write down the information you are to use during the presentation.

**Item 6**

Freshwater bodies like lakes and rivers have been heavily encroached on as a result of rapid urbanization which has impacted on the resource. On World Environment Day, the Ugandan government, particularly the National Environmental Management Authority (NEMA), wishes to raise public awareness among the citizens

The science club at your school has selected you to give a presentation on the occasion. On theme "FRESH WATER IS BASIC NECESITY".

**Task:**

Write the presentation you can use.

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**END**

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